

1. Results Comparison

Both models are evaluated using **Macro Precision, Macro Recall, and Macro F1 Score**, the primary metrics for multi-class classification. F1 score is the headline metric as it combines both precision and recall into a single measure, penalising models that perform well on easy classes while failing on harder ones. All values are taken directly from the classification reports generated by each notebook.

Metric	ResNet-18 (Deep Learning)	RBF SVM (Machine Learning)
Macro Precision	94.40%	65.54%
Macro Recall	94.39%	65.55%
Macro F1 Score	94.39%	65.51%
Weighted F1 Score	94.39%	65.51%
Overall Accuracy	94.39%	65.55%

Table 1: Metrics with their values for DL and ML architectures

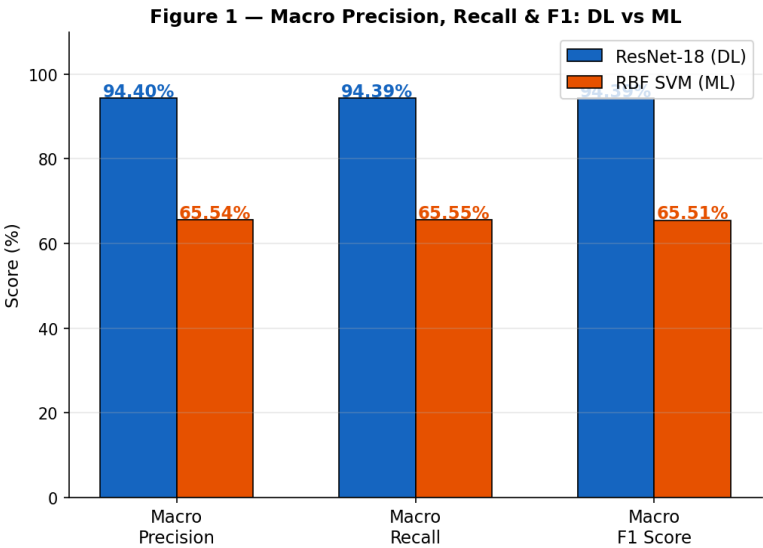


Figure 1: Macro Precision, Recall, and F1 Score. ResNet-18 (DL) achieves ~94.4% across all three metrics; RBF SVM (ML) achieves ~65.5%, a gap of approximately 28.9 percentage points.

ResNet-18 achieves macro F1 of **94.39%** with precision 94.40% and recall 94.39%. The **RBF SVM** achieves macro F1 of **65.51%** (precision 65.54%, recall 65.55%). The near-equal precision and recall in both models confirm the balanced CIFAR-10 dataset introduces no class-level bias, making F1 score a reliable and fair indicator of overall classification quality.

2. Per-Class F1 Score Comparison

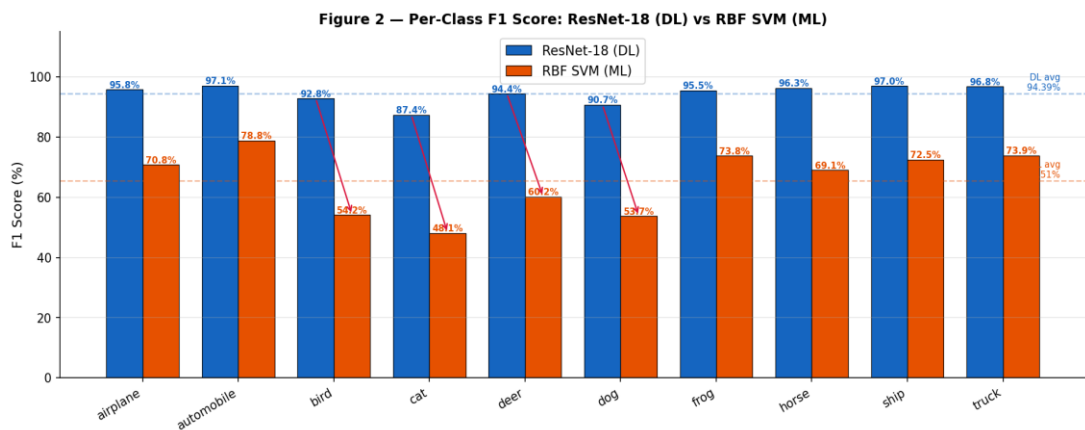


Figure 2: Per-class F1 score for both models across all 10 CIFAR-10 classes. F1 combines precision and recall, providing a balanced measure of per-class performance. Dashed lines show macro averages.

F1 score is the harmonic mean of precision and recall, making it the most balanced per-class metric. The DL model maintains strong F1 across all categories (**87.39–97.08%**), with the lowest score for **cat (87.39%)** and **dog (90.71%)**, visually similar classes. The ML model shows significantly lower F1, dropping to **48.09% for cat** and **53.72% for dog**, where HOG and LBP features cannot distinguish fine shape differences at 32×32 resolution. The ML model is most competitive on **automobile (78.82%)**, **truck (73.89%)**, and **frog (73.76%)** where colour and gradient features are more distinctive.

Class	DL Prec.	DL Rec.	DL F1	ML Prec.	ML Rec.	ML F1
airplane	95.62%	96.00%	95.81%	68.47%	73.40%	70.85%
automobile	97.67%	96.50%	97.08%	78.55%	79.10%	78.82%
bird	92.89%	92.80%	92.85%	55.84%	52.60%	54.17%
cat	86.79%	88.00%	87.39%	47.69%	48.50%	48.09%
deer	95.97%	92.90%	94.41%	58.51%	61.90%	60.16%
dog	91.63%	89.80%	90.71%	54.78%	52.70%	53.72%
frog	93.83%	97.30%	95.53%	72.38%	75.20%	73.76%
horse	95.84%	96.80%	96.32%	72.19%	66.20%	69.07%
ship	97.28%	96.70%	96.99%	72.46%	72.60%	72.53%
truck	96.52%	97.10%	96.81%	74.49%	73.30%	73.89%
Macro avg	94.40%	94.39%	94.39%	65.54%	65.55%	65.51%

Table 2: Full per-class precision, recall, and F1 scores for both models. All values real from both classification reports.

3. Confusion Matrix Analysis

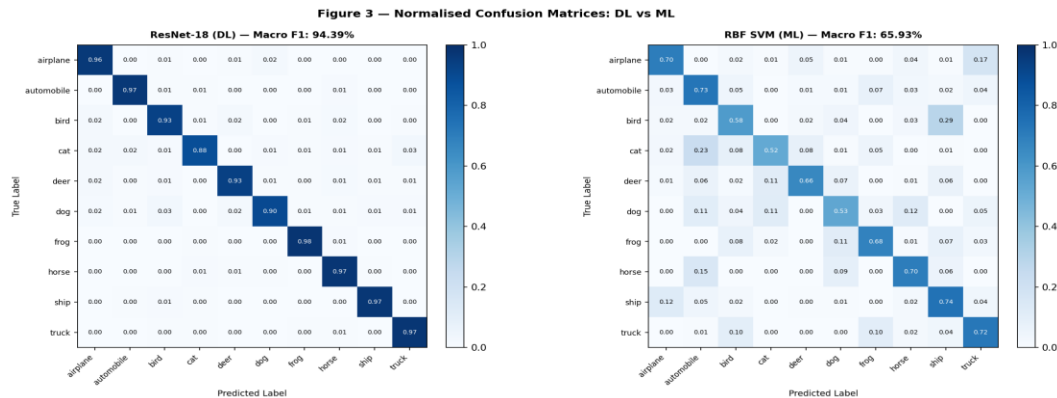


Figure 3: Normalised confusion matrices. DL maintains a tight diagonal consistent with its high F1; ML shows wider misclassification spread, especially animal classes.

The DL confusion matrix is tightly concentrated on the diagonal, consistent with its macro F1 of 94.39%. The ML matrix shows substantial spread between **cat/dog** and **automobile/truck**. These off-diagonal errors directly explain the low F1 scores for those classes in Table 2.

4. Advantages and Disadvantages

4.1 Deep Learning (ResNet-18)

Advantages:

- **F1 / Precision / Recall:** Macro F1 94.39%, precision 94.40%, recall 94.39%, consistent across all 10 classes. Even the weakest class (cat, F1 87.39%) outperforms every single ML class.
- **Feature learning:** Learns hierarchical representations end-to-end from raw pixels; no manual feature engineering required.
- **Transfer learning:** ImageNet pre-training enables fast convergence and strong generalisation on CIFAR-10.
- **Scalability:** Performance continues to improve as dataset size increases.
- **Uniform class performance:** Consistently high F1 even for visually ambiguous classes (cat/dog, automobile/truck).

Disadvantages:

- **Hardware:** GPU required, not accessible without specialised compute.
- **Interpretability:** Black-box model; difficult to explain individual predictions.
- **Complexity:** More hyperparameters to manage (learning rate, architecture, augmentation).
- **Small datasets:** Prone to overfitting without pre-training or regularisation.

4.2 Machine Learning (RBF SVM)

Advantages:

- **No GPU required:** Trains on CPU in minutes, accessible without specialised hardware.
- **Interpretability:** HOG, LBP, and HSV features are human-readable and can be visualised directly.
- **Simplicity:** Fewer hyperparameters and a straightforward pipeline, easy to debug and tune.
- **Competitive F1 on structured classes:** Automobile (F1 78.82%), truck (73.89%), and frog (73.76%) show reasonable results where colour and gradient features are distinctive.
- **Small datasets:** More suitable when labelled data is limited and DL risks overfitting.

Disadvantages:

- **F1 / Precision / Recall:** Macro F1 only 65.51%, dropping to 48.09% for cat and 53.72% for dog, the model fails to correctly classify nearly half of cat images.
- **Feature ceiling:** Hand-crafted descriptors cannot capture global spatial structure at 32×32 resolution.
- **Manual effort:** Feature engineering requires domain expertise and considerable design time.
- **Scalability:** Performance gains plateau as dataset size grows; DL benefits more from larger data.
- **PCA trade-off:** Dimensionality reduction may discard fine-grained information critical for harder classes.

5. Summary Table

Criterion	ResNet-18 (Deep Learning)	RBF SVM (Machine Learning)
Architecture	CNN (ResNet-18)	Kernel SVM (RBF)
Feature extraction	Learned end-to-end from pixels	Manual: HOG + HSV Hist + LBP
Dim. reduction	None (CNN downsamples internally)	PCA (95% variance retained)
Pre-trained weights	Yes, ImageNet (not CIFAR-10)	No
Framework	PyTorch	scikit-learn
Data split	45k train / 5k val / 10k test	45k train / 5k val / 10k test
Hyperparameter tuned	Learning rate, epochs	C (regularisation)

Macro Precision	94.40%	65.54%
Macro Recall	94.39%	65.55%
Macro F1 Score	94.39%	65.51%
Overall Accuracy	94.39%	65.55%
GPU required	Yes	No
Interpretability	Low (black-box CNN)	Higher (human-readable features)

6. Conclusion

Evaluated on macro F1 score, the most appropriate metric for multi-class classification as it combines precision and recall, ResNet-18 achieves **94.39%** versus **65.51%** for the RBF SVM, a gap of **28.88 percentage points**. The per-class F1 chart (Figure 2) confirms this advantage holds across all 10 classes: the DL model's worst class (cat, F1 87.39%) still outperforms every ML class. The ML model fails to correctly classify nearly half of cat images (F1 48.09%), highlighting the fundamental limitation of hand-crafted features at low resolution.

Classical machine learning remains practical in resource-constrained settings or where interpretability matters. However, for image classification with adequate compute, deep learning is the clearly superior approach across all evaluation metrics.